RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Mechanical Engineering

Date: 8th January 2024	**CIE – I**	**Max. Marks: 50**
Semester: 3	**UG**	**Duration: 90 Minutes**
Course Title: Materials Science for Engineers		**Course Code: ME232AT**

S No	Questions	M	BT	CO
1.	Describe all the primary and secondary bonds with examples.	10	2	1
2.	Explain energy bands for metals, Semiconductors, and insulators with a neat sketch.	10	3	1
3a.	Calculate Atomic packing factor for FCC unit cell.	04	4	1
3b.	Define the following with respect to solid materials: i) space lattice, ii) unit cell iii) Atomic Packing Factor.	06	1	1
4.	Explain properties and applications of ceramics and composites with examples.	10	2	1
5.	Explain the following defects: vacancy, edge dislocations, grain boundary.	10	2	1

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Department of Civil Engineering

Date	**09/10/2026**	**Maximum Marks**	**50**
Course Code	**CV232TA**	**Duration**	**90 Min**
Sem	**III**	**CIE – I**	
Environment and Sustainability (Basket Course)			

Sl. No	Questions	M	BT	CO
1.	Classify different types of eco system and explain any two briefly.	10	2	1
2.	Define solid waste management and list the objectives of solid waste management.	10	2	1
3.	i) The design population of a town is 15000. Determine the average daily demand, maximum hourly demand. Assume per capita consumption as 270 LPCD, Maximum daily demand as 1.8 times average daily demand and maximum hourly demand as 2.7 times annual average hourly demand.	05	3	1
	ii) A city generates 250 tonnes/day of municipal solid waste. The compacted density of waste in the landfill is 600 kg/m^3. Calculate the landfill volume required for one year.	05	3	1
4.	i) A municipality with a population of 5,00,000 generates solid waste at 0.55 kg/capita/day. The collection efficiency is 82%. Determine the total waste generated, waste collected and uncollected waste.	05	3	1
	ii) A drainage area consists of roofs, roads and lawns as shown below. If the rainfall intensity is 60 mm/h, determine the overall runoff coefficient and peak runoff.	05	3	1
5.	Write a note on: i) Rain water harvesting ii) Sustainable utilization of energy resources.	10	2	2

Surface	Area (ha)	Runoff coefficient, C
Roofs	6	0.90
Roads	5	0.80
Lawns	9	0.25

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	40	10	–	–	–	30	20	–	–	–

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Department of Civil Engineering

Date: 21/10/2024	**CIE – I (Quiz + Test)**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min + 20 Min**
Course Title: Environment and Sustainability (Basket Course)		**Course Code: CV232AT**

S No	Questions	M	BT	CO
	Part A			
1.	Distinguish between ecology and ecosystem.	2	1	1
2.	What is primary and secondary succession?	2	1	1
3.	Write any four methods of disposal of solid waste.	2	1	2
4.	What is the importance of safety in workplace?	2	1	2
5.	List any four environmental protection acts in India.	2	1	1
	Part B			
1a.	Briefly explain the structure and components of a ecosystem.	5	2	1
1b.	Materials flow in eco system is bi directional and Energy flow is unidirectional, justify.	5	2	1
2.	With a neat sketch explain the nitrogen cycle.	10	2	1
3.	Define solid waste management and list the objectives of solid waste management.	10	2	2
4.	Define Air Pollution? Classify and explain the different air pollutants and its effects on health and Environment.	10	2	2
5.	Briefly explain the following safety terminologies. i) Accident ii) Ergonomics iii) First aid iv) Hazard v) Safety inspection.	10	2	2

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Department of Civil Engineering

Date: 8th Jan 2024	**CIE – I**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min**
Course Title: Environment and Sustainability		**Course Code: CV232AT**

S No	Questions	M	BT	CO
1a	Define the terms i) Environment ii) Ecology iii) Bio-Diversity.	6	1	1
1b	Materials flow in eco system is cyclic and Energy flow is unidirectional, justify.	4	2	1
2	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition.	10	2	2
3a	Recall the concept of ecological succession and classify the same.	5	1	1
3b	Discuss the approaches of conserving the biodiversity.	5	2	1
4a	Highlight the salient features of various hotspots of biodiversity in India.	5	1	1
4b	Briefly discuss the causes, effects and preventive measures of water pollution.	5	2	2
5	Briefly describe the various Environmental protection acts in India.	10	2	2

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Department of Biotechnology

Date: 22nd October 2024	**Closed Book Offline Quiz & Test – 1**	**Max. Marks: 60**
Semester: III	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions (Quiz)	M	BT	CO
1	Covid 19 virus, will be categorized under which Bio safety level.	1	2	1
2	Mention the composition of Institutional Biosafety Committee.	1	2	2
3	Define pathogen, give example.	1	1	1
4	Mention the purpose of UV lamp in BSCs.	1	2	1
5	List out the working mechanisms of HEPA filters.	2	3	1
6	Expand the terms MERV and ULPA.	2	3	1
7	For the use of GMO at RVCE, what are the approvals required?	2	2	2

S No	Questions (Test)	M	BT	CO
1	Define Biohazard and mention the types of Biohazards with examples for each type.	10	2	1
2	Explain biosafety levels with a neat triangular diagram, with examples for each of the Biosafety level.	10	4	1
3	Illustrate on different types of Biosafety Cabinets, with the help of a neat diagram explain features and advantages of BSC-AII.	10	4	1
4a	Define GMO. Explain with a neat diagram the production of protein rich tomato using carrot gene.	7	2	2
4b	Explain importance and applications of GMOs in various fields.	3	2	2
5a	“Indian farmers, opposed the approval of Genetically Modified mustard seeds in the year 2022”, mention the reasons for opposition and justify the statement in your own words.	7	4	2
5b	Define Cartagena protocol and mention its important functions.	3	2	2

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Department of Biotechnology

Date: January 8, 2024	**Closed Book Offline Test – 1**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions (Test)	M	BT	CO
1	Elaborate on various bio hazards from nature.	10	2	1
2	List out and discuss biosafety levels with a neat triangular diagram.	10	4	1
3	Explain with a help of a neat diagram, biosafety cabinet B2.	10	4	1
4	Differentiate the bio safety cabinets I, II and III.	10	2	1
5	Give details on GMO and their applications in various fields.	10	2	2